

RENANCE

PETROLEUM TRAINING INSTITUTE, EFFURUN

MEC 422 — Past Questions & Materials

by Renance

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Paper 1 of 1 — MEC 422 Industrial Engineering Notes 2024 2025

Second Semester

Level HND2 • Second Semester • Handout

Industrial Engineering Function and Skills

Industrial engineering is that branch of engineering concerned with the planning, design, analysis, improvement, and installation of integrated systems of employees, materials, and equipment to produce a product or render a service.

In general, industrial engineering is concerned with the design of production and service systems. The industrial engineer analyses and specifies integrated components of people, machines, materials, and facilities to create efficient and effective systems that produce goods and services beneficial to mankind. Industrial engineers are the bridge between management goals and operational performance. They are expected to exert leadership in workplace education as well as analysis, design and implementation of systems and plans for operating those systems that bring value to the organization. Industrial engineers identify opportunities for improvement with equipment, materials, methods, layouts, and they develop plans

Operational Definition of Industrial and Systems Engineer

“An Industrial and Systems Engineer is one who is concerned with the design, installation, and improvement of integrated systems of people, material, information, equipment, and energy by drawing upon specialized knowledge and skills in the mathematical, physical, and social sciences, together with the principles and methods of engineering analysis and design to specify, predict, and evaluate the results to be obtained from such systems” (Womack and Jones 1996). This is the current and fairly traditional definition for ISE.

Functions and Responsibilities

Typical Industrial Engineering functional areas include:

Project Management

Responsibilities include the following:

- I. Develop the detailed work breakdown structure of complex activities and form them into an integrated plan
- II. Provide time-based schedules and resource allocations for complex plans or implementations
- III. Use project management techniques to perform Industrial Engineering analyses and investigations
- IV. Conduct facility planning and facility layout development of new and revised production plants and office buildings
- V. Form and direct both small and large teams that work towards a defined objective, scope & deliverables
- VI. Perform risk analysis of various project options and outcomes

Manufacturing, Production and Distribution

Responsibilities include the following:

- I. Participate in design reviews to ensure manufacturability of the product
- II. Determine methods and procedures for production distribution activity
- III. Create documentation and work instructions for production and distribution
- IV. Manage resources and maintain schedule requirements to meet required production and distribution schedules
- V. Process Optimization utilizing Simulation tools (Arena, etc)
- VI. Facilitate and Lead process improvement teams

Supply Chain Management

Responsibilities include the following:

- I. Manage Supplier relationships
- II. Managing and report on company Supplier Cost / Performance Indices to management
- III. Audit Suppliers and ensure supplier processes and procedures are being followed
- IV. Travel to suppliers to resolve issues
- V. Coordinate first article Inspections
- VI. Work with Outsource Manufacturers to ensure product quality, delivery and cost, is maintained

Productivity, Methods and Process Engineering

Responsibilities include the following:

- I. Define proper work methods for tasks
- II. Define appropriate processes for workflow activities
- III. Define key production measures
- IV. Define goals and data capture/analysis for key measures
- V. Perform root cause analysis to improve poor performing processes
- VI. Develop appropriate incentive plans for work tasks
- VII. Determine capacity requirements and subsequent investment options

Quality Measurement and Improvement

Responsibilities include the following:

- I. Resolve quality-related issues in all aspects of the business
- II. Work with design and production teams and outsource manufacturers to ensure product quality is maintained during the design and production phases
- III. Audit defined processes and procedures to ensure that they are being followed
- IV. Coordinate and Facilitate 3rd Party Quality Audits
- V. Provide refresher training on procedures for company personnel on Quality and process-related issues,
- VI. including the use of analytical tools and techniques such as SPC, Six Sigma, etc.
- VII. Manage and resolve issues with incoming material through the Receiving process

Program Management

Responsibilities include the following:

- I. Develop proposals for new programs
- II. Manage program/project teams to ensure program stays on schedule, on budget,

and meets performance expectations

III. Coordinate a matrix of team member across departments within an organization to ensure completion of project tasks

Ergonomics/Human Factors

Responsibilities include the following:

I. Ensure Human Factors Engineering is utilized in New Product Design

II. Ensure Human Factors Engineering disciplines are utilized in production setup and configuration

III. Ensure company Ergonomics policies are defined to minimize causes of employee injury and discomfort

Technology Development and Transfer

Responsibilities include the following:

I. Identify basic business problems requiring analysis

II. Determine if technology or process based solution best

III. Characterize problem, identify prospective providers/bidders and submit requests for proposals

IV. Evaluate bid responses, select successful bidder(s) and establish technical feasibility

V. Conduct small scale/medium scale tests to determine operational feasibility, implementation methods and training requirements

VI. Conduct enterprise wide implementation

VII. Transition support activities/responsibilities to long term business and technology owners

Strategic Planning

Responsibilities include the following:

I. Develop long range planning models, typically 5-10 years in scope

II. Model all areas affected by operation

III. Identify anticipated investment in plant, capacity, network, etc

IV. Tie to preliminary production cost, operational cost, sales forecasts

V. Develop preliminary financial impacts, including profitability and ROI

Management of Change

Responsibilities include the following:

I. Ensure that change programs are coordinated, support one another and move along the critical path

II. Create and maintain the imperative for the change, establish priorities and provide visible sponsorship for the change

III. Provide the skills, knowledge, processes, organization structure and tools required to deliver the change

IV. Ensure that the individuals involved buy into the change, actively support it and adopt their behaviour accordingly

Financial Engineering

Responsibilities include the following:

- I. Determine production costs using specific cost based methodology
- II. Develop budgets, forecasts for operating cost centres
- III. Measure actual performance vs budget goals and investigate variance
- IV. Develop capital and expense budgets for capacity expansion
- V. Perform cost analysis/justification for capital and expense expenditures
- VI. Perform 'make vs buy vs lease' analyses

Skills and Techniques Utilized by Industrial Engineers

- Benchmarking • Modelling & Testing • Strategic Planning
- Design of Experiments • Operations Auditing • Theory of Constraints
- Employee Involvement • Organizational Analysis • Time Studies
- Equipment Utilization • Pilot Programs • Work Sampling
- Flow Diagramming • Plant & Equipment

Layout

- Information & Data

Flow Diagramming • Project Management

- Interviewing for • Simulation

Information

- Six Sigma projects
- Lean Manufacturing
- Statistical Analysis

History of Industrial Engineering

Industrial engineering is one of the oldest branches of engineering, evolving alongside the rise of modern industry. It focuses on optimizing complex systems, processes, and organizations to improve efficiency, productivity, and quality.

Early Foundations (Pre-18th Century)

While the formal discipline of industrial engineering did not exist, many early practices reflected industrial engineering principles:

Ancient Egypt & Rome: Used techniques for project management and resource allocation in massive construction projects like pyramids and aqueducts.

Leonardo da Vinci (15th century): Designed machines and systems with considerations for efficiency and automation.

Industrial Revolution (18th–19th Century)

The Industrial Revolution laid the groundwork for industrial engineering:

1760–1840: Mechanization of manufacturing through steam power and factory systems.

Adam Smith (1776): Introduced the concept of division of labor in *The Wealth of Nations*—a key principle in industrial engineering.

Eli Whitney: Promoted interchangeable parts, which improved production efficiency.

Charles Babbage: Analyzed work processes and costs, publishing *On the Economy of Machinery and Manufactures* in 1832.

Scientific Management Era (Late 19th – Early 20th Century)

This period saw industrial engineering become more formalized as a discipline:

Frederick Winslow Taylor (Father of Scientific Management): Studied time and motion.

Promoted standardization and systematic training.

Frank & Lillian Gilbreth: Developed time and motion studies. Lillian pioneered ergonomics and human factors.

Henry Gantt: Created the Gantt chart—a tool for project scheduling and management.

Mid-20th Century: World Wars and Growth

World War I & II required better logistics, production planning, and operations research.

Operations Research (OR) emerged as a field to optimize military and industrial decisions.

Industrial engineering expanded into systems engineering, quality control, and human factors.

Late 20th Century – Present

Computers and Automation: Industrial engineers began using simulation, optimization algorithms, and statistical software.

Lean Manufacturing & Six Sigma:

Lean: Focus on waste reduction (popularized by Toyota Production System).

Six Sigma: Focus on reducing variation and improving quality.

Globalization: Increased focus on supply chain management, logistics, and international operations.

Today and the Future

Industrial engineering integrates data science, artificial intelligence, sustainability, and human-centered design to optimize modern systems. It applies to:

Manufacturing, Healthcare, Logistics, Finance, Tech and Service industries

Summary Timeline

Period Key Developments

Pre-1700s Early project and resource management in ancient civilizations

1760–1840 Industrial Revolution: factories, mechanization, division of labour

1880–1920 Scientific Management (Taylor, Gilbreths, Gantt)

1940s–60s Operations Research, Quality Control, WWII logistics

1980s–2000s Lean, Six Sigma, globalization

2000s–Today AI, big data, automation, sustainable systems

Industrial engineering has played a crucial role in industrial development by improving efficiency, productivity, and innovation across various sectors. Here are the key roles it has played:

1. Process Optimization

- Role: Designing and improving production processes.
- Impact: Increased output, reduced waste, and improved product quality.
- Example: Lean manufacturing and Six Sigma have roots in industrial engineering principles.

2. Resource Management

- Role: Efficient use of materials, machinery, and manpower.
- Impact: Cost savings and better allocation of limited resources.

- Example: Just-in-time (JIT) inventory systems to minimize holding costs.

3. Automation and Technology Integration

- Role: Incorporating automation, robotics, and computer systems into production.
- Impact: Higher speed, precision, and safety in industrial operations.
- Example: Implementing computer-integrated manufacturing (CIM) systems.

4. Quality Control and Assurance

- Role: Ensuring products meet specified standards and customer expectations.
- Impact: Enhanced brand reputation and reduced rework or recalls.
- Example: Statistical Process Control (SPC) to monitor manufacturing quality.

5. Supply Chain and Logistics Management

- Role: Streamlining the flow of goods, information, and finances.
- Impact: Faster delivery, lower costs, and improved customer satisfaction.
- Example: Optimization of global supply chain networks.

6. Human Factors and Ergonomics

- Role: Designing systems and workplaces that align with human capabilities.
- Impact: Reduced worker fatigue and injuries, leading to higher productivity.
- Example: Ergonomically designed assembly lines.

7. Project Management and Strategic Planning

- Role: Planning and executing industrial projects effectively.
- Impact: On-time delivery, within budget, and aligned with strategic goals.
- Example: Using Gantt charts and Critical Path Method (CPM) in factory construction.

8. Sustainability and Environmental Impact

- Role: Reducing energy usage and waste in production processes.
- Impact: More sustainable and eco-friendly industries.
- Example: Life Cycle Assessment (LCA) and green manufacturing.

Work System

Definition:

A work system is a structured arrangement of people, tasks, tools, technologies, information, and environment designed to produce specific outcomes or deliver services/products.

Components of a Work System:

1. People – Workers involved in performing tasks.
2. Tasks – Activities required to achieve objectives.
3. Tools & Technology – Machines, computers, software, etc.
4. Information – Data needed to perform the work.
5. Environment – Physical and organizational context.
6. Processes – Sequences and flows of work.
7. Outcomes – Products or services produced.

Purpose: To ensure all elements of work interact efficiently to achieve high productivity, quality, and employee satisfaction.

Work Design

Definition:

Work design (also called job design) refers to the process of organizing tasks, responsibilities, and workflows in a job to enhance performance, motivation, and well-being.

Key Elements of Work Design:

1. Task Variety – Different types of tasks included in the job.
2. Skill Variety – Range of skills required to perform the job.
3. Autonomy – Level of freedom and decision-making given to the worker.
4. Task Identity – Completing a whole piece of work.
5. Task Significance – Importance of the job's impact.
6. Feedback – Information on performance effectiveness.

Work Design Techniques:

- Job Rotation – Switching between tasks to reduce monotony.
- Job Enlargement – Increasing the number of tasks in a job.
- Job Enrichment – Adding depth by giving more control and responsibility.

Key Differences**Feature Work System Work Design**

Focus Entire system of work Individual job/task design

Broader (includes people, tech, info, Narrower (focuses on job/task

Scope

environment) structure)

Optimize system productivity & Improve job satisfaction, efficiency,

Objective

interaction and engagement

Manufacturing systems, service

Application Individual roles and job structures
systems

Why They Matter in Industrial Engineering

- Work system analysis helps identify inefficiencies and optimize resource use.
- Work design improves worker motivation, reduces fatigue, and boosts productivity.

Concept of Manufacturing

Manufacturing is the process of converting raw materials or components into finished goods using various tools, machines, human labour, and chemical or physical processes. It is a key part of the industrial sector and forms the foundation of economic development.

Manufacturing = Input (Raw Materials) → Process (Machines/Labor) → Output (Finished Goods)

It involves:

- Designing products
- Selecting raw materials
- Applying processes like machining, assembling, welding, or packaging

- Quality control and inspection

Purpose of Manufacturing

The main goals of manufacturing include:

Value Addition

- Transforms inexpensive raw materials into valuable products.
- Example: Turning iron ore into steel, then into car parts.

Meeting Consumer Demand

- Produces goods needed by society, from food to smartphones.

Economic Development

- Boosts GDP, employment, exports, and industrial growth.

Mass Production

- Enables production at scale, reducing cost per unit through economies of scale.

Innovation and Technological Advancement

- Drives research and development (R&D), automation, and new product creation.

Job Creation

- Provides employment to a wide range of skill levels—from labourers to engineers.

Self-Reliance

- Reduces dependency on imports and strengthens local industries.

Supply Chain Management (SCM)

Supply Chain Management (SCM) is the coordination and integration of all activities involved in the flow of goods, services, information, and finances — from raw material suppliers to the final customer.

It involves managing the entire network of:

- Suppliers
- Manufacturers
- Warehouses
- Distributors
- Retailers
- Customers

Objectives of Supply Chain Management

1. Deliver value to the customer
2. Ensure smooth flow of materials and information
3. Reduce costs and increase efficiency
4. Maintain product quality and availability
5. Achieve competitive advantage

Key Components of SCM

Component Description

Planning Forecasting demand and designing the supply chain structure

Sourcing Selecting suppliers and procuring raw materials

Manufacturing Converting raw materials into finished products

Delivery & Logistics Transporting goods to warehouses, distributors, or customers

Returns Handling product returns, recycling, or waste disposal

Flow in SCM

SCM manages three major flows:

1. Material Flow – Raw materials → Work-in-progress → Finished goods
2. Information Flow – Orders, forecasts, inventory levels
3. Financial Flow – Payments, credit, cost information

Production Planning and Control (PPC)

Production Planning and Control (PPC) is the process of planning, scheduling, and controlling all aspects of production to ensure goods are manufactured efficiently, cost-effectively, and on time.

Need for Production Planning and Control

1. Efficient Resource Utilization: Ensures optimal use of materials, labor, and machinery.
2. On-Time Delivery: Helps meet delivery deadlines by organizing work and avoiding delays.
3. Reduced Wastage: Minimizes waste of time, materials, and effort.
4. Cost Control: Helps reduce production costs through proper planning and coordination.
5. Inventory Management: Prevents overstocking or understocking of raw materials and finished goods.
6. Improved Productivity: Ensures a smooth flow of operations, reducing idle time and bottlenecks.
7. Customer Satisfaction: On-time delivery and consistent quality improve customer trust and satisfaction.

Objectives of Production Planning and Control

1. Ensure Availability of Inputs: Raw materials, tools, and labor must be ready when needed.
2. Maintain Steady Production Flow: Avoid interruptions and balance workloads.
3. Meet Delivery Schedules: Ensure products are completed and delivered on time.
4. Optimize Resource Use: Maximize output from available resources.
5. Control Production Costs: Reduce inefficiencies and improve cost-effectiveness.
6. Improve Quality: Coordinate quality checks to ensure product standards.
7. Flexibility to Demand Changes: Adjust production plans in response to market or order changes.

Functions of Production Planning and Control

1. Planning Function:
 - o Routing: Determining the path each product will take in the production process.
 - o Scheduling: Setting the time frame for each operation.

- o Loading: Assigning tasks to machines or workers based on capacity.

2. Control Function:

- o Dispatching: Issuing orders to start production as per schedule.
- o Follow-Up: Monitoring progress and ensuring schedules are followed.
- o Inspection: Checking the quality of raw materials and finished products.
- o Corrective Actions: Identifying and fixing delays, breakdowns, or quality issues.

Production Procedure

The production procedure refers to the step-by-step process followed to transform raw materials into finished goods. It typically includes:

Product Design and Development

- Creating product specifications and drawings.
- Testing prototypes and finalizing designs.

Process Planning

- Identifying production steps, machines, tools, and materials needed.

Routing

- Deciding the path or sequence of operations the product will follow.

Scheduling

- Allocating time for each operation or process step.

Loading

- Assigning work to specific machines or workers based on capacity.

Dispatching

- Starting actual production work with proper work orders and instructions.

Inspection and Quality Control

- Checking the product at various stages to ensure it meets standards.

Finishing and Packaging

- Final touches, packaging, and preparation for delivery.

Delivery and Feedback

- Shipping to customers and gathering feedback for improvements.

Manufacturing Methods

There are several methods of manufacturing based on production volume and product type:

Job Production

- Description: Customized, made-to-order products (e.g., shipbuilding, tailored clothing).
- Features: One-off production, skilled labour, high variety, low volume.

Batch Production

- Description: Producing a group (batch) of similar products (e.g., bakery items, furniture).
- Features: Intermediate volume, standardization within batches.

Mass (Flow) Production

- Description: Large-scale, continuous production of standardized products (e.g., cars,

electronics).

- Features: High volume, low variety, automation.

Continuous Production

- Description: Non-stop production for commodities like oil, cement, chemicals.
- Features: 24/7 operation, very high volume, minimal flexibility.

Project-Based Production

- Description: Used for large, one-time projects (e.g., construction, bridges).
- Features: Complex coordination, long duration, customized outputs.

Problems in Production Planning and Control

Despite its importance, PPC faces many challenges:

Inaccurate Forecasting

- Wrong demand predictions lead to overproduction or shortages.

Machine Breakdown

- Unexpected failures disrupt schedules.

Poor Coordination

- Lack of communication between departments affects workflow.

Unskilled Labor

- Incompetent workforce reduces productivity and quality.

Material Shortages

- Delays from suppliers or poor inventory management cause stoppages.

Rush or Urgent Orders

- Sudden priority changes disrupt existing schedules.

Inefficient Layout

- Poor factory design increases movement, delays, and bottlenecks.

Quality Issues

- Rework due to defects increases time and cost.

Facilities Layout

Facilities layout is the physical arrangement of machines, equipment, work areas, and resources within a manufacturing plant or service facility to ensure a smooth flow of materials, people, and information.

Objectives of a Good Layout

- Efficient material handling
- Reduced production time and cost
- Smooth workflow with minimal backtracking
- Flexibility for future expansion or modification
- Safe and ergonomic working environment

Problems in Facilities Layout

Common issues include:

Problem Description

Inefficient Material Excessive movement and transportation waste time
Flow and money.

Imbalanced workflows where one area slows down
Bottlenecks
the entire system.

Poor layout may waste floor space or cause
Space Utilization
overcrowding.

Longer distances between departments increase
High Handling Costs
handling time and cost.

Inability to adapt the layout to new products or
Lack of Flexibility
technologies.

Problem Description

Congested or poorly planned layouts can pose risks to
Safety Hazards
workers.

Types of Facilities Layout

Product (Line) Layout

- Used For: Mass production of standardized products.
- Arrangement: Machines are arranged in a line according to the sequence of operations.
- Example: Automobile assembly line.
- Pros: High efficiency, low handling cost.
- Cons: Inflexible to changes, one machine failure can stop the entire line.

Process (Functional) Layout

- Used For: Job or batch production with varied processes.
- Arrangement: Machines grouped by similar functions (e.g., all drills in one area).
- Example: Machine shop.
- Pros: Flexibility, better machine utilization.
- Cons: Complex material flow, higher handling cost.

Fixed-Position Layout

- Used For: Large, heavy, or immovable products.
- Arrangement: Workers and equipment move around the product.
- Example: Shipbuilding, aircraft manufacturing.
- Pros: Suitable for large projects.
- Cons: Planning and coordination are complex.

Cellular Layout

- Used For: Medium-volume, varied product lines.
- Arrangement: Machines are grouped into cells that produce similar items.

- Example: Manufacturing cells for parts families.
- Pros: Reduced movement, better control.
- Cons: Requires careful grouping and layout design.

Combination/Hybrid Layout

- Used For: Plants that produce both standardized and custom products.
- Arrangement: Mix of product and process layouts.
- Example: Large factories like electronics plants.
- Pros: Versatile and efficient.
- Cons: Complex to design and manage.

Line Balancing

Line balancing is the process of assigning tasks to workstations so that each has an equal amount of work and there is no delay in the production line.

Objectives

- Minimize idle time of workers and machines
- Reduce cycle time (time per unit)
- Maximize production efficiency
- Ensure smooth workflow

Steps in Line Balancing

- List tasks required to complete the product.
- Determine time required for each task.
- Identify task sequence (precedence diagram).
- Calculate cycle time (available time / desired output).
- Assign tasks to stations so total task time is close to cycle time.
- Evaluate efficiency and adjust as needed.

Problems with Poor Line Balancing

- Idle machines/workers
- Bottlenecks and delays
- Lower throughput
- Higher labour cost per unit

Job Design

Job design is one of those aspects of managing organizations that is so commonplace that it often goes unnoticed. Most people realize the importance of job design when an organization is being built, and production processes are being developed. Some even recognize the importance of job design when changes are taking place in organizational structures and processes. However, few people realize that job design may be affected when organizations grow, retrench, or reorganize, when managers use their discretion in the assignment of miscellaneous tasks on a day-to-day basis, or when the people in the jobs or their managers change. Fewer yet realize that job design change can be used as an intervention to enhance important organizational goals.

Many different aspects of an organization influence job design, including an organization's

structure, technology, processes, and environment. These influences impose constraints on how jobs are designed. However, considerable discretion often exists in the design of jobs in many organizational situations. The job (defined as a set of tasks performed by a worker) is a convenient unit of analysis in developing new organizations or changing existing ones.

Several distinctions are useful to clarify the terminology. One distinction is between a task and a job. A task is a set of actions performed by a worker who transforms inputs into outputs through the use of tools, equipment, or work aids. The actions of the task may be physical, mental, or interpersonal. On the other hand, a job is an aggregation of tasks assigned to a worker (Gael 1983; U.S. Department of Labour 1972). When the same set of tasks is performed by more than one worker, those workers are said to have the same job.

It is often useful to distinguish among positions, jobs, and occupations. A position is the set of tasks performed by one worker (e.g., the specific industrial engineering position held by employee X). A job is a set of similar positions (e.g., the industrial engineering positions in manufacturing in a particular company). The tasks performed in a given position are usually a combination of tasks that are common to all workers in that job and of tasks that are unique to that position. The unique tasks are sometimes a function of organizational requirements (e.g., different product or equipment) and sometimes a function of the disposition of the particular worker (e.g., different strengths or interests). An occupation is a collection of similar jobs (e.g., all industrial engineering jobs across companies). Job design usually focuses, by definition, on the job level. Differences in design between positions are assumed to be small and are often ignored. This may not be the case in all situations, however. And there can be great differences in design across jobs within an occupation.

Many of the personnel and productivity problems in industry may be the direct result of the design of jobs. Job design can have a strong influence on a broad range of important efficiency and human resource outcomes:

- Productivity
- Quality
- Job satisfaction
- Training time
- Intrinsic work motivation
- Staffing
- Error rates
- Accident rates
- Mental fatigue
- Physical fatigue
- Stress
- Mental ability requirements
- Physical ability

requirements

- Job involvement
- Absenteeism
- Medical incidents
- Turnover
- Compensation rates

Meaning and Definition of Work Measurement:

Work measurement is concerned with the determination of the amount of time required to perform a unit of work. Work measurement is very important for promoting productivity of an organization. It enables management to compare alternate methods and also to do initial staffing. Work measurement provides basis for proper planning.

Since it is concerned with the measurement of time it is also called 'Time Study'. The exact examination of time is very essential for correct pricing. To find the correct manufacturing time for a product, time study is performed. To give competitive quotations, estimation of accurate labour cost is very essential. It becomes a basis for wage and salary administration and devising incentive schemes.

Work measurement has been defined by British Standard Institution as, "The application of techniques designed to establish the time for a qualified worker to carry out a specified job at a defined level of performance". This time is called standard or allowed time. Time study may also be defined as "the art of observing and recording the time required to do each detailed element of an industrial operation".

Objectives of Work Measurement:

1. To compare the times of performance by alternative methods.
2. To enable realistic schedule of work to be prepared.
3. To arrive at a realistic and fair incentive scheme.
4. To analyse the activities for doing a job with the view to reduce or eliminate unnecessary jobs.
5. To minimise the human effort.
6. To assist in the organisation of labour by daily comparing the actual time with that of target time.

Uses of Work Measurement:

1. Work measurement is used in planning work and in drawing out schedules.
2. Work measurement is used to determine standard costs.
3. Work measurement is used as an aid in preparing budgets.
4. It is used in balancing production lines for new products.
5. Work measurement is used in determining machine effectiveness.
6. To determine time standards to be used as a basis for labour cost control.
7. To establish supervisory objectives and to provide a basis for measuring supervisory efficiency.

8. To determine time standards to be used for providing a basis for wage incentive plans.

Techniques of Work Measurement:

Work measurement is investigating and eliminating ineffective time. It not only reveals the existence of ineffective time. But it can be used to set standard times for carrying out the work so that ineffective time does not evolve later. It will be immediately found out by the increased standard time. For the purpose of work measurement, work may be regarded as repetitive work and non-repetitive work.

The principal techniques of work measurement are classified under the following heads:

1. Time Study: Time study refers to the ascertainment of the time needed to carry out a unit of work. In this method, observation and recording of time is necessary for undertaking each unit of an operation are done, with a view to ascertaining, the actual time, in which the work can be accomplished.
2. Work Sampling: A work measurement method, in which the work of several employees is sampled randomly, at periodic intervals, to ascertain the proportion of total operations, of a specific activity.
3. Pre-determined Motion Time System: In PMTS method, basic times are set up for basic human motions. Such time values are used to compute the time required by the job for its completion, with fixed standard. It is a new and improved version of motion study.
4. Analytical Estimating: This method of time measurement is used to ascertain the time values for the tasks, that are long and not repetitive in nature.

Steps Involved in Work Measurement

- I. Divide jobs into elements
- II. Observe and record each element, any of the work measurement techniques.
- III. Set up unit time values, by extending observed time into normal time for each unit. This can be done by applying rating factor.
- IV. Evaluate relaxation allowance and add the same to the normal time, for each element to get the work content.
- V. Ascertain the frequency of occurrence of each element in the job, then multiply the work content to it. After that total the times to reach the work content of the job.
- VI. Add contingency allowance, wherever required, to get the standard time for performing the job.
- VII. Work measurement is helpful in evaluating the labour cost. Further, gives information with respect to the estimation of tenders, assessment of delivery schedule and fixation of the selling price.

ERGONOMICS

Definition:

Ergonomics can be defined simply as the study of work. More specifically, ergonomics is the science of designing the job to fit the worker, rather than physically forcing the worker's body to fit the job. Adapting tasks, workstations, tools, and equipment to fit the worker can

help reduce physical stress on a worker's body and eliminate many potentially serious, disabling work related musculoskeletal disorders (MSDs). Ergonomics draws on a number of scientific disciplines, including physiology, biomechanics, psychology, anthropometry, industrial hygiene, and kinesiology.

"Ergonomics applies information about human behaviour, abilities and limitations and other characteristics to the design of tools, machines, tasks, jobs and environments for productive, safe, comfortable and effective human use" (McCormick and Saunders 1993).

A number of factors play a role in Ergonomics; these include body posture and movement (sitting, standing, lifting, pulling and pushing), and environmental factors (noise, lighting, temperature, humidity). (Dul et al 1993).

How can ergonomics and human factors improve health and safety?

Applying ergonomics to the workplace can:

- reduce the potential for accidents.
- reduce the potential for injury and ill health.
- improve performance and productivity.

Taking account of ergonomics and human factors can reduce the likelihood of an accident.

For example, in the design of control panels, consider:

- the location of switches and buttons – switches that could be accidentally knocked on or off might start the wrong sequence of events that could lead to an accident;
- expectations of signals and controls – most people interpret green to indicate a safe condition. If a green light is used to indicate a 'warning or dangerous state' it may be ignored or overlooked;
- information overload – if a worker is given too much information, they may become confused, make mistakes, or panic. In hazardous industries, incorrect decisions or mistaken actions have had catastrophic results.

Ergonomics can also reduce the potential for ill health at work, such as aches, pains and damage to the wrists, shoulders and back, noise-induced hearing loss and work-related asthma. Consider the layout of controls and equipment – they should be positioned in relation to how they are used. Place those used most often where they are easy to reach without the need to stoop, stretch or hunch. Making sure protective measures such as extraction hoods or respirators are easy and comfortable to use means they are more likely to be effective at reducing exposure to hazardous substances.

THE IMPACT OF ERGONOMICS ON WORKPLACE DESIGN

The goal of Ergonomics is to provide maximum productivity with minimal cost; in this context cost is expressed as the physiological or health cost to the worker. In a workplace setting there are seldom a large number of tasks that exceed the capabilities of most of the work force. There may be jobs that will include a specific task that requires extended reaches or overhead work that cannot be sustained for long periods, by using Ergonomic principles to design these tasks; more people should be able to perform the job without the risk of injury.

Ergonomics has already been defined and its primary focus is on the design of work activity that suits the person in that it takes account of their capabilities and limitations. Matching the requirements of a job with the capabilities of the worker is the approach to be adopted in order to reduce the risks of musculoskeletal injuries resulting from handling materials.

Proactive Ergonomics emphasises the prevention of work related musculoskeletal disorders through recognising, anticipating and reducing risk factors in the planning stages of new systems of work or workplaces. In effect, to design operations that ensures proper selection and use of tools, job methods, workstation layouts and materials that impose no undue stress and strain on the worker. Additional costs are incurred in redesigning or modifying work processes therefore it is more cost effective to reduce risk factors at the design stage.

A proactive approach to Ergonomics will ensure that:

- Designers will receive training in ergonomics and have appropriate information and guidelines regarding risk reduction
- Decision-makers planning new work processes should have knowledge of Ergonomics principles that contribute to the reduction or elimination of risk.
- Design strategies emphasise fitting job demands to the capabilities and limitations of workers. For example, for tasks requiring heavy materials handling, use of mechanical assist devices to reduce the need for manual handling would be designed into the process
- Other aspects of design should be considered including load design, layout of the workplace to allow for ease of access when using mechanical aids and eliminating unnecessary lifting activities.

ERGONOMIC PRINCIPLES THAT CONTRIBUTE TO GOOD WORKPLACE DESIGN

The goal for the design of workplaces is to design for as many people as possible and to have an understanding of the Ergonomic principles of posture and movement which play a central role in the provision of a safe, healthy and comfortable work environment. Posture and movement at work will be dictated by the task and the workplace, the body's muscles, ligaments and joints are involved in adopting posture, carrying out a movement and applying a force. The muscles provide the force necessary to adopt a posture or make a movement. Poor posture and movement can contribute to local mechanical stress on the muscles, ligaments and joints, resulting in complaints of the neck, back, shoulder, wrist and other parts of the musculoskeletal system.

ERGONOMICS: POINTS TO REMEMBER

- Ergonomics applies information about human behaviour, abilities and limitations and other characteristics to the design of tools, machines, tasks, jobs and environments for productive, safe, comfortable and effective human use
- The goal for the design of workplaces is to design for as many people as possible and to have an understanding of the Ergonomic principles of posture and movement which play a central role in the provision of a safe, healthy and comfortable work environment

- Risk Assessment of manual handling activities and Display Screen equipment workstations should always take account of Ergonomic Principles. The Schedules outline the appropriate factors that must be considered as part of the risk assessment process.

PRODUCTIVITY AND PRODUCTIVITY IMPROVEMENT

Concept

Productivity is the ratio between output and input. It is a quantitative relationship between what we produce and what we have spent to produce. Productivity is nothing but reduction in wastage of resources like men, material, machine, time, space, capital etc. It can be expressed as human efforts to produce more and more with less and less inputs of resources so that there will be maximum distribution of benefits among maximum number of people. Productivity denotes relationship between output and one or all associated inputs.

On the other hand, according to International Labour Organisation (ILO), productivity is the ratio between the volume of output as measured by production indices and the corresponding volume of labour input' as measured by production indices and the corresponding volume of labour input as measured by employment indices. This definition applies to an enterprise, industry or an economy.

The productivity of a certain set of resources (input) is therefore the amount of goods or services (output) which is produced by them. Land and building materials, machines, manpower (labour), technology etc. are the resources at the disposal of a manufacturing company. Therefore higher (improved) productivity means that more is produced with the same expenditure of resource i.e. at the same cost in terms of land, materials, machine, time or labour, alternatively, it means same amount is produced at less cost in terms of land, materials, machine time or labour that is utilized.

Importance of Improved Productivity:

If the level of output is increased faster than that of input, productivity will increase. Conversely, productivity will be increased if the level of input is decreased faster than that of output. Also, an organization may realize a productivity increase from producing more output with the same level of input.

Finally, producing more output with a reduced level of input will result in increased productivity.

Any of these scenarios may be realized through improved methods, investment in machinery and technology, improved quality, and improvement techniques and philosophies such as just-in-time, total quality management, lean production, supply chain management principles, and theory of constraints.

A firm or department may undertake a number of key steps toward improving productivity.

These are the steps to productivity improvement:

- Develop productivity measures for all operations; measurement is the first step in managing and controlling an organization.

- Look at the system as a whole in deciding which operations are most critical; it is over-all productivity that is important.
- Develop methods for achieving productivity improvement, such as soliciting ideas from workers (perhaps organizing teams of workers, engineers, and managers), studying how other firms have increased productivity, and re-examining the way work is done.
- Establish reasonable goals for improvement.
- Make it clear that management supports and encourages productivity improvement. Consider incentives to reward workers for contributions.
- Measure improvements and publicize them.
- Don't confuse productivity with efficiency. Efficiency is a narrower concept that pertains to getting the most out of a given set of resources; productivity is a broader concept that pertains to use of overall resources. For example, an efficiency perspective on mowing the lawn given a hand mower would focus on the best way to use the hand mower; a productivity perspective would include the possibility of using a power mower.

Therefore, it is essential to know the importance of improved productivity in manufacturing company/ organization. Thus, importance of productivity can be summarized as follows:

- I. Productivity is a key to prosperity. Rise in productivity results in higher production which has direct impact on standard of living. It reduces cost per unit and enables reduction in sale price. It increases wages for workers and increased profit for organisation. Higher demand creates more employment opportunities.
- II. Improved productivity leads to economic growth and social progress. Higher productivity helps to reduce cost per piece which make product available at cheaper rate. Thus, it is beneficial for consumers. Low price increases demand of the product which in turn increases profit of the organisation. Higher profit enables organisation to offer higher dividend for shareholders. It increases export and increases foreign exchange reserves of a country.
- III. Improved productivity requires elimination of waste in all forms. It is necessary to eliminate wastage in raw material, wastage of time in case of men and machinery, wastage of space etc. to improve productivity. Several techniques like work study, statistical quality control, inventory control, operation research, value analysis etc. Are used to minimise wastage of resources.
- IV. Improvement in productivity is important for country like ours because it can minimise level of poverty and unemployment.

TECHNIQUES FOR MEASUREMENT OF PRODUCTIVITY

Productivity has been defined as the ratio of output to input. An increase in productivity means an increase in output that is proportionally greater than increase in input.

Productivity may be measured either on an aggregate basis or individual basis. On aggregate basis, output is compared with all inputs taken (added) together. This is called as total productivity. On individual basis, output is compared with any one of the input factor and

this is called as partial productivity or factor productivity.

Output / Input

Output / Input =

Output / Input

Output / Input = Output / Input

Output / Input + Output / Input + Output / Input

This index measures the productivity of the entire organisation with use of all resources. It is a way of evaluating efficiency of entire plant or firm.

the challenge of productivity has become a challenge of measurement. Productivity is difficult to measure and can only be measured indirectly, that is, by measuring other variables and then calculating productivity from them. This difficulty in measurement stems from the fact that inputs and outputs are not only difficult to define but are also difficult to quantify.

Any productivity measurement system should produce some sort of overall index of productivity. A smart measurement program combines productivity measurements into an overall rating of performance. This type of system should be flexible to accommodate changes in goals and policies over time. It should also can aggregate the measurement systems of different units into a single system and be able to compare productivity across different units.

Total Productive maintenance (TPM):

TPM is keeping machines in good working condition through systematic maintenance of equipment so that they fail less frequently, and production process continues without interruption.

Just-In-Time:

JIT is a management philosophy aimed at eliminating waste from every aspect of manufacturing and its related activities. The term JIT refers to producing only what is needed, when it is needed and in needed quantity. The aim of JIT in a factory is to reduce lead times, minimise inventory, reduce the defect rate to zero and accomplish all of the above at minimum cost. There are three essential ingredients to effective manufacturing excellence through JIT:

- (i) JIT manufacturing techniques that aims to promote a rapid response to customer demand while minimising inventory
- (ii) a total quality culture to pursue excellence in both the product and every area of the business, including customer service, purchasing, order taking, accounting, maintenance, design, etc.; and
- (iii) people or employee involvement in the development of the organisation through its culture and its manufacturing and other business processes.

Work Standards:

Work standards represent the best way of doing a job and it consists of set of documented policies, rules, directives and procedures established by the management for all major operations to enable employees to perform their jobs without errors and to enable

management to minimize variations in output, quality, work-in-progress and cost. Three elements of standardized work are task time, work sequence and standard work-in-progress. The key steps in implementing work standards are

- (i) identify the key issues in the current process
- (ii) map the process
- (iii) improve the process
- (iv) implement the process and
- (v) sustain the process.

The PDCA/SDCA Cycle:

The PDCA (Plan-Do-Check-Act) cycle is an endless improvement cycle which demands that each team Plans (establish a target for improvement through action plan), Does (implement the plan), Checks (monitor and evaluate effects), Acts (Standardises new procedures or set goals for further improvement). Since every process is initially unstable, it must be stabilize using SDCA (Standardise-Do-Check - Act) cycle. Thus, a SDCA cycle must precede every PDCA cycle to consolidate gains into current process before raising the standard threshold.

Statistical Process Control (SPC):

Statistical Process Control is the application of statistical techniques to control a process and eliminate process variations due to assignable causes. Statistical process control requires operators to do periodically sampling of the quality of their own output, enter data into control charts, analyse the trend, and decide for themselves when to shut down the process and when to make adjustments / corrections to the process to prevent defects.

Work Study as a Productivity Improvement Technique:

Work study is an important management tool to achieve higher productivity. It is related to human work, method of doing work and standard of performance. The survival of any organization is dependent on use of latest technology and efficient methods of production. To improve efficiency of production it needs effective utilization of plant, equipment and labour. This can be achieved by using work study which studies method and evaluate the performance. It divides work into smaller elements, studies it, and rearranges it to get same or greater efficiency at reduced cost. International Labour Organization (ILO) defines work study as the technique of method study and work measurement employed to ensure the best possible use of human and material resources in carrying out a specified activity. It is also a management service based on method study and work measurement used in examination of human work leading to investigation of all the resources that effect efficiency and economy of situation to affect improvement. Further ILO states that work study is to minimize cost either by designing the work for high productivity or by improving productivity in existing work through improvements in current methods by reducing ineffective and wasted time. Therefore, it can be said that it is a direct means of raising the productivity. It is most frequently used to increase the amount of production from a given quantity of resources with little or no further capital investment and hence work study has direct relation to productivity improvement. According to B.S Glossary, " Work study is a generic term for those techniques,

particularly method study and work measurement, which are used in the examination of human work in all its contexts, and which lead systematically to the investigation of all the factors which affect the efficiency and economy of the situation being reviewed, in order to effect improvement"6. From this definition we can say that, work study is direct means of rising or improving the productivity as work study is used to examine human work that affects efficiency which in turn affects improvement. Work study therefore has direct relationship with the productivity. It is frequently used to increase the amount produced from a given quantity of resources with little or no further investment. Productivity of an enterprise is affected by various factors such as Labour, Material, Land, Machine, Capital, Technology, product and Management. One of the effective ways of raising productivity in the long run is the development of new process and installation of modern plant and equipment. But this approach requires heavy capital investment. Further, to improve the productivity using advanced technology may affect the efforts aimed at expanding employment opportunities. On the other hand, work study aims to improve productivity through the systematic analysis of existing operations, processes and work methods so as to improve their efficiency with little or no extra capital expenditure. To achieve this improvement it is extremely important to apply work study continuously and throughout the organization. Thus, International Labour organization has clearly shown to what extent there is direct relationship of work study in raising the productivity.